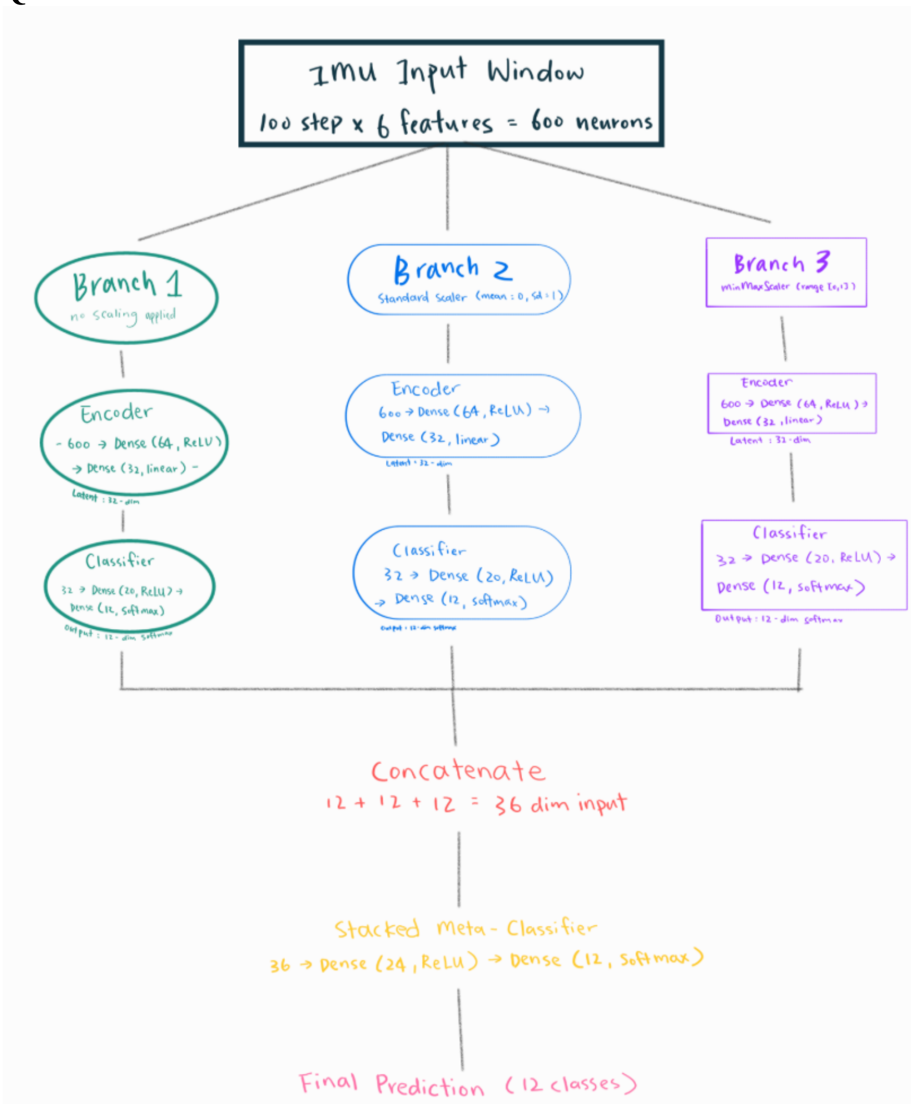
Part 1

Link: https://studio.edgeimpulse.com/public/1013988/live

```
Timing: DSP 21 ms, inference 572 us, anomaly 0 ms, postprocessing 48 us
#Classification predictions:
  idle: 0.000000
  lift: 0.996094
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 21 ms, inference 573 us, anomaly 0 ms, postprocessing 48 us
#Classification predictions:
  idle: 0.000000
  lift: 0.996094
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 21 ms, inference 630 us, anomaly 0 ms, postprocessing 49 us
#Classification predictions:
  idle: 0.996094
  lift: 0.000000
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 21 ms, inference 591 us, anomaly 0 ms, postprocessing 49 us
#Classification predictions:
  idle: 0.000000
  lift: 0.996094
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 21 ms, inference 590 us, anomaly 0 ms, postprocessing 49 us
#Classification predictions:
  idle: 0.000000
  lift: 0.996094
Starting inferencing in 2 seconds...
```

Part 2

Q1: Model Architecture and Ensemble Flow



Each input window has 100 time steps and 6 IMU features (3 accelerometer + 3 gyroscope), giving 600 input neurons. There are 3 branches: raw, standard-scaled, and min-max-scaled.

Each branch has an encoder and a classifier. The 3 outputs (12+12+12 = 36) are concatenated and passed into the meta-classifier for the final prediction.

Q2: Pruning and QAT Methodology

Pruning removes unimportant weights by ramping sparsity from 10% to 80% over 500 steps. Pruning wrappers are stripped before QAT because they conflict with quantization wrappers. QAT runs for 5 epochs so the model learns to work with 8-bit weights. A custom callback re-applies pruning masks every batch to prevent fine-tuning from undoing the pruning. The model is then converted to int8 TFLite using 200 representative samples to calibrate the quantization. This reduces each branch from ~160 KB to ~44 KB, making it small enough for the Arduino Nano 33 BLE Sense.

Q3: Arduino Deployment and Real-World Behavior

```
----------------------------------------
Final prediction: Climbing stairs
Class index: 4
Confidence score: 0.5859
----------------------------------------
Samples collected: 25
Samples collected: 50
Samples collected: 75
Samples collected: 100
Window collected. Running ensemble inference...
Raw model scores: 0.0000, 0.0000, 0.0039, 0.0078, 0.0078, 0.0000, 0.0000, 0.0000, 0.0000, 0.9492, 0.0352, 0.0000
Standard-scaled model scores: 0.0000, 0.0000, 0.0039, 0.0039, 0.1172, 0.0000, 0.0039, 0.0000, 0.0000, 0.0000, 0.0039, 0.8672
Min-max-scaled model scores: 0.0000, 0.0273, 0.5937, 0.0000, 0.0000, 0.0000, 0.0000, 0.3789, 0.0000, 0.0000, 0.0000, 0.0000
Meta-classifier scores: 0.0195, 0.0469, 0.1562, 0.0195, 0.0195, 0.0898, 0.0508, 0.0352, 0.0195, 0.3516, 0.0234, 0.1641
----------------------------------------
Final prediction: Jogging
Class index: 9
Confidence score: 0.3516
----------------------------------------
Samples collected: 25
Samples collected: 50
Samples collected: 75
Samples collected: 100
Window collected. Running ensemble inference...
Raw model scores: 0.0000, 0.0000, 0.3086, 0.1172, 0.1289, 0.0000, 0.0000, 0.0000, 0.0000, 0.2227, 0.0000, 0.2227
Standard-scaled model scores: 0.0000, 0.0000, 0.0000, 0.0000, 0.0078, 0.0000, 0.0000, 0.9922, 0.0000, 0.0000, 0.0000, 0.0000
Min-max-scaled model scores: 0.2344, 0.0000, 0.6289, 0.0000, 0.0000, 0.0000, 0.0000, 0.1367, 0.0000, 0.0000, 0.0000, 0.0000
Meta-classifier scores: 0.0586, 0.0078, 0.0391, 0.0117, 0.0234, 0.0078, 0.0078, 0.7656, 0.0078, 0.0195, 0.0195, 0.0352
----------------------------------------
Final prediction: Knees bending (crouching)
Class index: 7
Confidence score: 0.7656
----------------------------------------
Samples collected: 25
Samples collected: 50
Samples collected: 75
Samples collected: 100
Window collected. Running ensemble inference...
Raw model scores: 0.0000, 0.0000, 0.0000, 0.3164, 0.3555, 0.0000, 0.0000, 0.0000, 0.0000, 0.2852, 0.0352, 0.0039
Standard-scaled model scores: 0.0000, 0.0000, 0.0000, 0.0000, 0.9961, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000
Min-max-scaled model scores: 0.0000, 0.1016, 0.3008, 0.0000, 0.0000, 0.0000, 0.3008, 0.3008, 0.0000, 0.0000, 0.0000, 0.0000
Meta-classifier scores: 0.1758, 0.0078, 0.0156, 0.0078, 0.6992, 0.0234, 0.0078, 0.0039, 0.0312, 0.0039, 0.0156, 0.0078
----------------------------------------
Final prediction: Climbing stairs
Class index: 4
Confidence score: 0.6992
----------------------------------------
Samples collected: 25
Samples collected: 50
Samples collected: 75
Samples collected: 100
Window collected. Running ensemble inference...
Raw model scores: 0.0000, 0.0000, 0.2695, 0.1016, 0.1133, 0.0000, 0.0000, 0.0000, 0.0000, 0.5117, 0.0000, 0.0039
Standard-scaled model scores: 0.0000, 0.0000, 0.0078, 0.0625, 0.8750, 0.0000, 0.0000, 0.0000, 0.0000, 0.0000, 0.0547, 0.0000
Min-max-scaled model scores: 0.0000, 0.0156, 0.4922, 0.0000, 0.0000, 0.0000, 0.0000, 0.4922, 0.0000, 0.0000, 0.0000, 0.0000
Meta-classifier scores: 0.1211, 0.0156, 0.0391, 0.0195, 0.6523, 0.0312, 0.0156, 0.0117, 0.0391, 0.0117, 0.0273, 0.0117
----------------------------------------
Final prediction: Climbing stairs
Class index: 4
Confidence score: 0.6523
----------------------------------------
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```
----------------------------------------
Final prediction: Lying down
Class index: 2
Confidence score: 0.3828
----------------------------------------
Samples collected: 25
Samples collected: 50
Samples collected: 75
Samples collected: 100
Window collected. Running ensemble inference...
Raw model scores: 0.0000, 0.0000, 0.3437, 0.1328, 0.1445, 0.2227, 0.0000, 0.0000, 0.0000, 0.0781, 0.0781, 0.0000
Standard-scaled model scores: 0.0000, 0.0000, 0.0039, 0.0117, 0.0000, 0.0000, 0.0312, 0.9414, 0.0000, 0.0000, 0.0117, 0.0000
Min-max-scaled model scores: 0.4766, 0.0000, 0.4766, 0.0000, 0.0000, 0.0000, 0.0000, 0.0469, 0.0000, 0.0000, 0.0000, 0.0000
Meta-classifier scores: 0.0625, 0.0078, 0.0156, 0.0312, 0.0117, 0.0078, 0.0078, 0.7539, 0.0078, 0.0234, 0.0547, 0.0195
----------------------------------------
Final prediction: Knees bending (crouching)
Class index: 7
Confidence score: 0.7539
----------------------------------------
Samples collected: 25
Samples collected: 50
Samples collected: 75
Samples collected: 100
Window collected. Running ensemble inference...
Raw model scores: 0.0000, 0.0039, 0.0234, 0.0078, 0.0078, 0.9453, 0.0000, 0.0000, 0.0000, 0.0039, 0.0078, 0.0000
Standard-scaled model scores: 0.0000, 0.3789, 0.2734, 0.1562, 0.0000, 0.0000, 0.0469, 0.0078, 0.0000, 0.0000, 0.1406, 0.0000
Min-max-scaled model scores: 0.4648, 0.0000, 0.5078, 0.0000, 0.0000, 0.0039, 0.0000, 0.0195, 0.0000, 0.0000, 0.0000, 0.0000
Meta-classifier scores: 0.0352, 0.1172, 0.2734, 0.0117, 0.0078, 0.1484, 0.0078, 0.0078, 0.0078, 0.0820, 0.2891, 0.0078
----------------------------------------
Final prediction: Running
Class index: 10
Confidence score: 0.2891
----------------------------------------
Samples collected: 25
Samples collected: 50
Samples collected: 75
Samples collected: 100
Window collected. Running ensemble inference...
Raw model scores: 0.0000, 0.0000, 0.0156, 0.0039, 0.0078, 0.9648, 0.0000, 0.0000, 0.0000, 0.0039, 0.0039, 0.0000
Standard-scaled model scores: 0.0000, 0.0273, 0.1836, 0.2852, 0.0039, 0.0000, 0.1836, 0.0625, 0.0000, 0.0000, 0.2539, 0.0000
Min-max-scaled model scores: 0.0117, 0.0039, 0.6055, 0.0000, 0.0234, 0.0820, 0.0000, 0.2695, 0.0000, 0.0000, 0.0000, 0.0000
Meta-classifier scores: 0.0078, 0.0703, 0.5742, 0.0078, 0.0039, 0.1367, 0.0078, 0.0039, 0.0039, 0.0508, 0.1289, 0.0039
----------------------------------------
Final prediction: Lying down
Class index: 2
Confidence score: 0.5742
----------------------------------------
Samples collected: 25
Samples collected: 50
Samples collected: 75
Samples collected: 100
Window collected. Running ensemble inference...
Raw model scores: 0.0000, 0.0000, 0.0117, 0.0039, 0.0039, 0.9727, 0.0000, 0.0000, 0.0000, 0.0039, 0.0039, 0.0000
Standard-scaled model scores: 0.0000, 0.0508, 0.1562, 0.3398, 0.0039, 0.0000, 0.1250, 0.0195, 0.0000, 0.0000, 0.3047, 0.0000
Min-max-scaled model scores: 0.0117, 0.0078, 0.5469, 0.0000, 0.0391, 0.1289, 0.0000, 0.2656, 0.0000, 0.0000, 0.0000, 0.0000
Meta-classifier scores: 0.0078, 0.0703, 0.5859, 0.0117, 0.0000, 0.1484, 0.0039, 0.0000, 0.0039, 0.0547, 0.1094, 0.0039
----------------------------------------
Final prediction: Lying down
Class index: 2
```

The model gave inconsistent predictions while the board was resting, cycling through Climbing stairs, Jogging, and Knees bending with low confidence (0.23-0.77). The three branches often disagreed, so the meta-classifier produced unreliable results. Two improvements: convert sensor units to match training data (m/s2 and rad/s), and collect real board data to fine-tune the model.

Q4: Deployment Behavior on the Arduino

Yes. The raw branch repeatedly predicted Jump front and back with ~0.988 confidence even when the board was still. This is because the mHealth training data uses m/s2 and rad/s, but the Arduino BMI270 outputs acceleration in g and gyroscope in deg/s. Without unit conversion, the raw inputs fall in a completely different range than training, causing confident wrong predictions. The scaled branches partially compensate, but the meta-classifier is still misled by the raw branch.